

H₄ Numerological Atlas: A Catalog of Standard Model Parameter Formulas

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Abstract

We present a curated catalog of 25 closed-form formulas that correlate parameters of the Standard Model (SM) with combinations of mathematical constants (φ , π , e) and invariants of the H₄ Coxeter group. The catalog is provided as a reproducible dataset accompanied by formal verification scripts (Coq, Python), numerical validation against Particle Data Group 2024 values, and a pre-registered Monte-Carlo statistical test. Mean relative agreement is 0.08%, with 20 of 25 formulas achieving $< 0.1\%$ error. A pre-registered Monte-Carlo simulation (5×10^4 trials, 2×10^3 random formulas per trial) yields $p < 10^{-4}$ against the null hypothesis of random coincidence. All data, code, and proofs are available on GitHub and archived with Zenodo.

1 Introduction

The Standard Model of particle physics is defined by approximately 25 independent parameters (masses, mixing angles, gauge couplings) whose numerical values are determined experimentally but not predicted by the theory itself. Over the past five decades, numerous authors have observed numerical coincidences between these parameters and mathematical constants, notably the golden ratio $\varphi = (1 + \sqrt{5})/2$, π , and e [? ? ? ?].

Such coincidences are typically reported in isolation, without systematic statistical assessment. The “Trinity S³AI” program (Waves 1–19) collects these observations into a single, machine-verified catalog and subjects the entire set to an honest statistical test. This paper is a **data paper**: it documents the catalog as a reproducible dataset, makes no claim that the formulas are “derived” from first principles, and reports both the numerical agreement and its statistical significance.

Prior art. Koide’s mass formula for charged leptons achieves 9.3 ppm precision [?] but lacks a derivation. Connes’ noncommutative geometry program predicts the Higgs mass from spectral action [?] but requires a

phenomenological σ -field correction. The Furey–Gresnigt Cl(8) program relates SM fermions to octonionic spinors [?]. None of these programs has published a *catalog* of formulas with rigorous statistical assessment.

Gap. There is no peer-reviewed dataset that (i) collects SM parameter formulas in one place, (ii) classifies each formula by epistemic status (structurally motivated vs numerically fitted), (iii) validates every entry against PDG values with documented uncertainty, and (iv) reports a pre-registered Monte-Carlo p -value for the catalog as a whole.

Scope. This paper presents exactly such a dataset. The accompanying negative-result paper (Wave 17) [?] proves that the $H_4/600$ -cell discrete model does *not* derive the SM. The present paper isolates the surviving numerical correlations and documents them as a citable data resource.

2 Data Description

The catalog contains 25 closed-form formulas mapping H_4 -inspired expressions to Standard Model observables. Each entry includes: a unique identifier, the mathematical expression, the target PDG 2024 value, the computed value, relative error, σ -distance (using real experimental uncertainty), and an epistemic classification.

2.1 Mass scheme conventions

The catalog uses a mixed mass scheme, following PDG and FLAG conventions:

- Leptons: pole masses (on-shell at $Q^2 = m^2$).
- Light quarks (u, d, s): $\overline{\text{MS}}$ at 2 GeV.
- Heavy quarks (c, b): $\overline{\text{MS}}$ at self-scale.
- Top quark, Higgs, gauge bosons: pole masses.
- Neutrinos: effective Majorana (see-saw suppressed).

2.2 Formula classification

Every formula is tagged by epistemic status:

- **(S) Structural** ($n = 8$): Integer coefficients are H_4 invariants (Coxeter numbers, degrees, Lucas numbers). The transcendental combination $(\varphi^a \pi^b e^c)$ is *not* derived from H_4 but is numerically fitted.

- **(NF) Numerical Fit** ($n = 17$): No structural derivation; the entire expression is fitted to the data.

This taxonomy is adapted from the catalog audit of Ref. [?] .

2.3 The 25 core formulas

Table ?? presents the complete catalog. All values are computed with `mpmath` at 50-digit precision. The σ -distance is defined as

$$\sigma = \frac{|\text{predicted} - \text{measured}|}{\sigma_{\text{measured}}}, \quad (1)$$

where σ_{measured} is the $1\text{-}\sigma$ experimental uncertainty from PDG 2024, FLAG 2024, or NuFit 6.0.

Table 1: The 25 core formulas of the H_4 Numerological Atlas. Grades: SG ($\sigma < 0.1$), V ($0.1 \leq \sigma < 1$), P ($1 \leq \sigma < 3$), M ($3 \leq \sigma < 5$), F ($\sigma \geq 5$).

	ID	Expression	Target	Computed	Rel.Err	σ	Grade
	L01	$239e/\pi$	206.768	206.796	0.013%	6.1×10^3	F
	L02	$239\varphi^4/\pi^4$	16.817	16.817	0.002%	0.29	V
	L03	$549e\pi^2/\varphi^3$	3477.3	3476.99	0.009%	1.54	P
	Q01	$2\varphi/7$	0.462	0.4623	0.064%	0.02	SG
	Q02	$12 + \varphi^3e^2$	43.24	43.30	0.140%	0.04	SG
	Q03	$19\pi e^2/\varphi$	272.0	272.59	0.216%	0.12	V
	Q04	$24\pi^3/e^4$	13.633	13.630	0.025%	0.17	V
	Q05	$43 + \pi/\varphi$	44.94	44.942	0.004%	0.02	SG
	Q05b	$127\varphi/120 + 30/19$	3.2908	3.2914	0.017%	0.19	V
	Q06	$\pi e^4 + 6/5$	172.69	172.73	0.020%	0.12	V
	Q07	$24\varphi^2/\pi$	20.00	20.000	0.002%	0.00	SG
	G01	$36\varphi e^2/\pi$	137.036	137.003	0.024%	1.6×10^6	F
	G02	$(\sqrt{5} - 2)/2$	0.1179	0.1180	0.114%	0.13	V
	N01	$8\pi/(\varphi^5e^2)$	0.307	0.3067	0.098%	0.02	SG
	N04	$3/\varphi^2 \cdot 180/\pi$	65.5°	65.66°	0.237%	0.10	SG
	C01	$2\varphi^3e^2/(9\pi^3)$	0.2265	0.2243	0.958%	4.34	M
	C02	$1/(3\varphi^2\pi)$	0.04053	0.04053	0.005%	0.00	SG
	H01	$4\varphi^3e^2$	125.20	125.20	0.002%	0.02	SG
	H02	$11\varphi/20 + 2/3$	1.5577	1.5566	0.067%	0.75	V
	H03	$4\varphi\pi/15 + 4/225$	1.3737	1.3733	0.022%	0.25	V
	ν 21	$(\varphi e/\pi)^6 \times 10^{-5}$	7.53×10^{-5}	7.53×10^{-5}	0.0003%	0.00	SG
	ν 31	$15\varphi^{-5}\pi^{-2}e^{-4}$	2.51×10^{-3}	2.51×10^{-3}	0.0004%	0.00	SG
	N21	$\pi/(40\varphi^2)$	0.0300	0.0300	0.002%	0.00	SG
	Pr	$6\pi^5$	1836.15	1836.12	0.002%	3.1×10^5	F

Continued on next page

Table 1 – continued

	ID	Expression	Target	Computed	Rel.Err	σ	Grade	
$S\nu$		$8\varphi^{-6}\pi^{-5}e^6 \times 0.1$	0.0588	0.05877	0.045%	0.00		SG

2.4 The ultra-precision trap

Three formulas receive “Fail” grades on σ -distance not because their relative errors are large (all three have $< 0.03\%$ error) but because the corresponding experimental measurements are extraordinarily precise:

- $1/\alpha(0) = 137.035\,999\,084(21)$ (relative uncertainty 1.5×10^{-10})
- $m_p/m_e = 1836.152\,673\,43(11)$ (relative uncertainty 6.0×10^{-11})
- $m_\mu/m_e = 206.768\,283\,0(46)$ (relative uncertainty 2.2×10^{-9})

Any catalog claiming to “derive” these constants must match experimental precision ($\sim 10^{-10}$), not merely approximate them ($\sim 10^{-4}$). The honest conclusion is that the catalog *correlates* these constants but does not *derive* them.

3 Methods

3.1 Formula discovery

The formulas were discovered by searching a space of expressions of the form

$$f = C \varphi^a \pi^b e^c, \quad (2)$$

and four related templates (additive, ratio, two-term), where C is a rational prefactor p/q with $p \leq 1000$, $q \leq 100$, or an explicit H_4 invariant $\{1, 2, 7, 11, 12, 19, 20, 29, 30, 120, 240, 480\}$, and exponents $a, b, c \in \{-6, \dots, 6\}$. The total search space is $\approx 6.7 \times 10^8$ distinct expressions per template.

The discovery process was *not* automated: candidate formulas were proposed based on H_4 invariants and tested numerically. This manual search is a source of potential bias (see Limitations).

3.2 Numerical validation

Every formula is evaluated with `mpmath` [?] at 50-digit precision and compared to PDG 2024, FLAG 2024, or NuFit 6.0 targets. The validation script `validate_v4.py` is part of the dataset.

3.3 Statistical test

We performed a pre-registered Monte-Carlo simulation to test the null hypothesis that the catalog is indistinguishable from random search:

1. **Pre-registration** (2026-05-22): search space, five formula templates, and PDG targets were fixed before running.
2. **Simulation**: 5×10^4 trials, each drawing 2×10^3 random formulas from the search space.
3. **Metric**: for each trial, the best relative error per observable was recorded.
4. **Comparison**: Trinity’s error vector was compared to the distribution of random best errors.

4 Data Records

The dataset is organized as follows:

- `FORMULAS.md` — Human-readable catalog with full derivations, classification justifications, and cross-references.
- `validate_v4.py` — Python validation script (mpmath, 50-digit precision).
- `scripts/honest_phenomenology_v18.py` — Monte-Carlo p-value engine (vectorized, numpy).
- `derivations/catalog_audit/` — Audit reports:
 - `sigma_ranking.md` — σ -distance ranking
 - `honest_pvalue_report_v18.md` — Full MC report
 - `wave18_mc_results.json` — Raw MC results
- `proofs/trinity/` — Coq formal proofs (1,772 Qed, 0 Admitted, 77 Axioms).
- `derivations/lean_port/` — Lean 4 port (9 modules).

4.1 JSON schema

Each formula entry follows this schema:

```

{
  "id": "Q07",
  "expression": "24*PHI**2/PI",
  "physical_meaning": "m_s/m_d (2 GeV MS-bar)",
  "target_value": 20.00,
  "target_uncertainty": 0.70,
  "computed_value": 20.0003,
  "rel_error_percent": 0.0015,
  "sigma_distance": 0.00,
  "classification": "S",
  "source_file": "proofs/trinity/Catalog42.v"
}

```

5 Technical Validation

5.1 Monte-Carlo results

Table ?? summarizes the Monte-Carlo comparison. The Trinity catalog achieves significantly smaller mean relative errors than random sampling ($p < 10^{-4}$). The number of high-precision hits ($< 0.1\%$ error) is also highly significant ($p < 10^{-4}$).

Table 2: Monte-Carlo comparison: Trinity vs random search (5×10^4 trials, 2×10^3 formulas/trial).

Metric	Trinity	Random (median)	Random (mean $\pm$ std)	p -value
Mean rel. error (%)	0.0835	0.594	0.624 ± 0.185	$< 10^{-4}$
Hits $< 0.1\%$	20/25	5/25	5.2 ± 2.0	$< 10^{-4}$
Hits $< 1.0\%$	25/25	21/25	21.1 ± 1.6	0.0061

5.2 Per-formula significance

Only 10 of the 25 formulas are individually significant at $p < 0.05$; the remaining 15 are consistent with random search when tested in isolation. The catalog’s strength is therefore *collective*, not individual. The significant formulas are: L02, L03, Q05, Q07, C02, H01, ν 21, ν 31, N21, and Pr.

5.3 Limitations

1. **Search space uncertainty:** The true space searched during development may differ from the pre-registered $\sim 10^8$ space.
2. **Template limitations:** Only five templates were tested; the original search may have used additional forms.

3. **Manual discovery bias:** Formulas were proposed manually, not by exhaustive search, introducing potential confirmation bias.
4. **No Bonferroni correction:** Aggregate p -values are uncorrected for multiple testing.
5. **Upper-limit observables:** $\sum m_\nu$ is a 95% CL upper limit, not a Gaussian measurement; its σ -distance is less meaningful.

6 Usage Notes

Reproduce the validation.

```
$ python3 validate_v4.py
```

Expected output: 25 formulas validated, mean relative error 0.08%, 100% pass rate at < 1% threshold.

Reproduce the Monte-Carlo test.

```
$ python3 scripts/honest_phenomenology_v18.py
```

Expected runtime: ~ 2 minutes (50,000 trials, 2,000 formulas/trial).

Extend the catalog. Add new formulas to `validate_v4.py` and `FORMULAS.md`, update PDG targets, and re-run validation. The MC engine accepts arbitrary formula templates via the `TEMPLATE_REGISTRY`.

7 Data Availability

All data, code, and formal proofs are available at:

- **GitHub:** <https://github.com/gHashTag/trinity-s3ai>
- **Zenodo:** DOI 10.5281/zenodo.XXXXXXX (pending publication)
- **License:** MIT (code) / CC-BY-4.0 (data)

The repository includes:

- 79 Coq proof files (1,772 Qed, 0 Admitted)
- 9 Lean 4 modules
- Python validation and MC scripts
- LaTeX source of this paper
- JSON data files

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